

Wi-Fi Embedded Webcam

M. KATHIRESAN, A. ROBSON BENJAMIN
AND C. VIJAYAN

This project is built around an MRF24WBOMA Wi-Fi module, a PIC microcontroller and a camera module using the TCP-IP protocol. The camera captures live video from a remote location, which is transmitted wirelessly through the Internet and can be viewed on a PC or laptop.

The design uses easily-available low-cost components and a serial peripheral interface (SPI). Microchip's recently introduced MRF24WBOMA (see Fig. 4) is an inexpensive Wi-Fi chip that uses SPI communication. Block diagram of the Wi-Fi embedded webcam is shown in Fig. 2.

Circuit and working

The circuit diagram for MRF24WBOMA-based Wi-Fi embedded webcam is shown in Fig. 3.

As shown in the block diagram



Fig. 1: Author's prototype

(Fig. 2), the power supply for the circuit includes 9V AC, a rectifier, a 7805 fixed regulator and an LM317 adjustable regulator. This supply drives the main circuit, which includes the Wi-Fi module, the 74HCT125 buffer, the PIC microcontroller and the MAX232 driver.

Power supply. The 9V derived from transformer X1 is rectified by rectifier module BR1, filtered by capacitor C1 and given to 7805 IC, which is a fixed 5V regulator. The LM317 adjustable regulator is configured to give a fixed 3.2V meant for the MRF24WBOMA Wi-Fi module. LED1 indicates the presence of power in the circuit.

MRF24WBOMA module. The MRF24WBOMA is a low-power, 2.4GHz, IEEE Std. 802.11-compliant, surface mount module with all associated RF components—crystal oscillator, bypass and bias passives with integrated MAC, baseband, RF and power amplifier, and built-in hardware support for

AES and TKIP. The MRF24WBOMA module is approved for use with the integrated PCB meander antenna. The PCB antenna is fabricated on the top copper layer and covered with solder mask.

The module interfaces to the PIC microcontroller via a serial slave SPI interface for interrupt, hibernate, reset, power and ground signals. The module normally runs on a single supply voltage of 3.3V, though here we have used a 3.2V supply. It also supports optional JTAG and serial debug for testing. The debug port operates at 3.3V and requires a level shifter for operation with RS232 devices.

The MRF24WBOMA module is designed for use with Microchip's TCP/IP software stack. The software stack

PARTS LIST

Semiconductors:

- IC1 - MRF24WBOMA Wi-Fi module
- IC2 - 74HCT125 quad-buffer
- IC3 - PIC18F4620 microcontroller
- IC4 - MAX232 RS232 driver
- IC5 - 7805, 5V regulator
- IC6 - LM317 adjustable regulator
- BR1 - 1A bridge rectifier module
- LED1, LED2 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 680-ohm
- R2 - 330-ohm
- R3, R11, R15-R17 - 220-ohm
- R4, R14 - 10-kilo-ohm
- R5-R9, R12, R13 - 4.7-kilo-ohm
- R10 - 1-mega-ohm
- VR1 - 10-kilo-ohm pot meter

Capacitors:

- C1 - 470uF, 25V electrolytic
- C2 - 0.1u ceramic disc
- C3 - 1uF, 25V electrolytic
- C4, C5 - 33pF ceramic disc
- C6-C9 - 0.1uF ceramic disc

Miscellaneous:

- X1 - 230V AC primary to 9V, 500mA secondary transformer
- LCD1 - 16x2 LCD module
- DB9 - 9-pin D-type COM port connector
- X_{cr}1 - 10MHz crystal oscillator
- WEBCAM - uCAM serial JPEG camera

Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	3.2V
TP3	Train of pulses when circuit is on
TP4	Train of pulses when circuit is on
TP5	Train of pulses when camera is connected

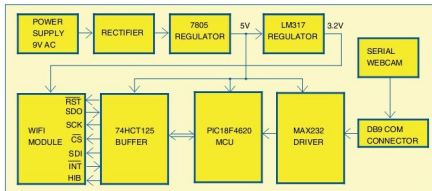


Fig. 2: Block diagram of the Wi-Fi embedded webcam